BOE-604	**NON-CONVENTIONAL ENERGY RESOURCES**	**L T P 3 0 0**	**3 Credits**

Course Objectives:

CO1	To understand the various forms of conventional energy resources.
CO2	To learn the present energy scenario and the need for energy conservation.
CO3	To understand the application of wind energy as alternate source of energy.
CO4	To outline division aspects and utilization of renewable energy sources for both domestics and industrial application.
CO5	To analyze the environmental aspects of renewable energy resources.

MODULE 1

Introduction
Various non-conventional energy resources- Introduction, availability, classification, relative merits and demerits.Solar Cells: Theory of solar cells. solar cell materials, solar cell array, solar cell power plant, limitations.
Solar Thermal Energy:
Solar radiation, flat plate collectors and their materials, applications and performance, focussing of collectors and their materials, applications and performance; solar thermal power plants, thermal energy storage for solar heating and cooling, limitations.

MODULE II

Geothermal Energy:

Resources of geothermal energy, thermodynamics of geo-thermal energy conversion-electrical conversion, non-electrical conversion, environmental considerations.

Magneto-hydrodynamics (MHD): Principle of working of MHD Power plant, performance and limitations. Fuel Cells: Principle of working of various types of fuel cells and their working, performance and limitations.

MODULE III

Thermo-electrical and thermionic Conversions: Principle of working, performance and limitations. Wind Energy: Wind power and its sources, site selection, criterion, momentum theory, classification of rotors, concentrations and augments, wind characteristics. performance and limitations of energy conversion systems. Bio-mass:

Availability of bio-mass and its conversion theory. Ocean Thermal Energy Conversion (OTEC): Availability, theory and working principle, performance and limitations.
Wave and Tidal Wave: Principle of working, performance and limitations.
Waste Recycling Plants.

. **Course Outcomes:**

CO1	To understand the need of energy conversion and the various methods of energy storage.
CO2	To explain the field applications of solar energy.
CO3	Identify Winds energy as alternate form of energy and to know how it can be tapped.
CO4	To understand the Geothermal &Tidal energy, its mechanism of production and its applications.
CO5	Illustrate the concepts of Direct Energy Conversion systems & their applications.

Text Books:
1. Raja etal, "Introduction to Non-Conventional Energy Resources" Scitech Publications.
2. John Twideu and Tony Weir, "Renewal Energy Resources" BSP Publications, 2006.
3. M.V.R. Koteswara Rao, " Energy Resources: Conventional & Non-Conventional " BSP Publications,2006.
4. D.S. Chauhan,"Non-conventional Energy Resources" New Age International.
5. C.S. Solanki, "Renewal Energy Technologies: A Practical Guide for Beginners" PHI Learning.